



## Choosing the Number of Clusters (K) Transcript

Remember in the last video we were discussing about how do you find the optimal number of clusters in k-means clustering. So there are various different types of methods which people use. The two of them, the two most important of them are known as ELBO curve and SILHOU distance.

So ELBO curve is, remember I told you about within cluster sum of squares, this has to be a minimum because within a cluster the distance between data points has to be low. If that is there, then you are actually looking for minimum of within clusters sum of squares with number of clusters. So what do you actually do is, you start plotting within cluster sum of squares with  $k$  is equals to 1 clusters with  $k$  is equals to 2 clusters, you find that WCSS value for  $k$  is equals to 1 cluster.

So you will run the algorithm with one cluster, you will run the algorithm with two clusters, you will run the algorithm with three clusters and so on and you will plot this curve wherein on your y-axis is within cluster sum of squares and on your x-axis is the number of that you have taken. So once you plot it at a point wherein if the plot looks like ELBO curve, the joint wherein the slope of it becomes zero, the entire graph or the entire function, it looks like zero, that is the point which you choose as the number of clusters, the optimal number of clusters because after this point if you think of there is very minimal change in within cluster sum of squares. So we treat the optimal number of clusters as the point wherein the ELBO curve or let's say when you look at ELBO, you have a joint there, so you have to choose that joint as the number of clusters.

So similarly we have different types of methods, SILHOUT is also a same type of method which you have as ELBO method, it's just that you use different metrics to quantify it. I'm not going to get into the details of it because it is very similar to what you have as ELBO method and both of them operate in a very similar fashion. So it is because it is a heuristics-based method wherein you have some idea about how to cluster distances.

-----END-----